gaussian_elimination

```python
# https://www.geeksforgeeks.org/dsa/gaussian-elimination/
MAXN = 100

def partial_pivot(A, n):
    for i in range(n):
        pivot_row = i
        for j in range(i + 1, n):
            if abs(A[j][i]) > abs(A[pivot_row][i]):
                pivot_row = j
        if pivot_row != i:
            A[i], A[pivot_row] = A[pivot_row], A[i]
        for j in range(i + 1, n):
            factor = A[j][i] / A[i][i]
            for k in range(n + 1):
                A[j][k] -= factor * A[i][k]

def back_substitute(A, n):
    x = [0]*n
    for i in reversed(range(n)):
        sum_val = sum(A[i][j] * x[j] for j in range(i + 1, n))
        x[i] = (A[i][n] - sum_val) / A[i][i]
    return x

if __name__ == "__main__":
    n = 3
    A = [[3.0, 2.0, -4.0, 3.0],
         [2.0, 3.0, 3.0, 15.0],
         [5.0, -3, 1.0, 14.0]]
    partial_pivot(A, n)
    x = back_substitute(A, n)
    print("Solution for the system:")
    for i in range(n):
        print(f"{x[i]:.6f}")

# 3.000000
# 1.000000
# 2.000000
```

goodbye

```
print("goodbye world")
```

hello

```
print("hello world")
```